

New feature for efficiency , lout and eff% note 2026/8/21

I would like to add a new feature for new efficiency formula, lout formula and derive $\eta\%$.

1. New efficiency formula.

There are 4 existing formula right now. They are

$$\eta = \frac{(1 - |s_{11}|^2) \times |S_{21}|^2}{|S_{21}|^2 \cdot (1 - |S_{22}|^2) / |1 - S_{22}^2|^2}$$
$$\eta_{ABCD} = \frac{P_L}{P_{in}}$$

I would like to add 5th formula as:

$$D_Z = Z_{11}(Z_L + Z_{22}) - Z_{12}Z_{21}$$
$$\eta = \frac{Re(Z_L)|Z_{21}|^2}{Re[D_Z(Z_L + Z_{22})^*]} \%$$

This result should be the same as 4th formula. But I would like to double confirm the result are the same. So I would like to add this as 5th formula.

Please make 5th formula as default going forward.

2. New lout formula.

I have one lout formula in tab lout now. I would like to add one more as:

$$D_Z = Z_{11}(Z_L + Z_{22}) - Z_{12}Z_{21}$$
$$I_{out} = \frac{Z_{21}V_1}{D_Z} \text{ (A)}$$

The result of new formula should be the same as existing formula. But I would like to double confirm it.

Please make the current iout formula as pull down manu just like efficiency tab, so user could select formula 1 and 2. Please leave new formula as default.

3. I like the way you handle mouse in tab Display. It will show blue color when user mouse hover over cell. Please implement the same way for other table such as X-Y Table, Phase Magnitude, Contour, Impedance, Zpar, ABCD Matrix, dZ, Reflect Coefficient, Efficiency and lout.

4. New tab called Vpp next to lout

The format just like lout, but different formula. Here is the formula:

$$D_Z = Z_{11}(Z_L + Z_{22}) - Z_{12}Z_{21}$$

$$V_{RMS} = \frac{Z_L Z_{21} V_1}{D_Z}$$

$$V_{pp} = 2\sqrt{2}V_{RMS}$$

Just like lout way, please add Input power(W) for user to enter forward power and assume input impedance is 50ohm to calculate correct V_1 . The table format should be just like lout format.

5. Smith Chart tab

Please leave Smith Chart tab as default tab when user just start this App.

Please leave Efficiency as default selection instead of X-Y Table right now.

6. New tab called 'Derive η formula' next to Vpp tab

Add below note in the tab with scroll bar on right side for user to move up and down.

How to derive simple term of η formula?

Single Z-Parameter Formula for Match Efficiency

1. Starting definitions

The Z-parameter convention is:

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

Both (I_1) and (I_2) are defined as currents flowing **into** the two-port network.

For the connected load, define (I_L) as the current flowing **out of the Match and into the load**:

$$I_2 = -I_L$$

and:

$$V_2 = Z_L I_L$$

2. ABCD-based efficiency equation

The ABCD matrix is:

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ I_L \end{bmatrix}$$

With:

$$V_2 = Z_L I_L$$

we have:

$$V_1 = (AZ_L + B)I_L$$

$$I_1 = (CZ_L + D)I_L$$

The real power entering the Match is:

$$P_{in} = \text{Re}\{V_1 I_1^*\}$$

Therefore:

$$P_{in} = \text{Re}\{(AZ_L + B)(CZ_L + D)^*\} |I_L|^2$$

The real power delivered to the load is:

$$P_L = \text{Re}\{Z_L\} |I_L|^2$$

Thus, the internal Match efficiency is:

$$\eta_{Match} = \frac{\text{Re}\{Z_L\}}{\text{Re}\{(AZ_L + B)(CZ_L + D)^*\}}$$

The arbitrary current (I_L) cancels, so you do not need to know the actual load current.

3. Substitute Z-parameters into the ABCD equation

The Z-to-ABCD conversion is:

$$A = \frac{Z_{11}}{Z_{21}}$$

$$B = \frac{Z_{11}Z_{22} - Z_{12}Z_{21}}{Z_{21}}$$

$$C = \frac{1}{Z_{21}}$$

$$D = \frac{Z_{22}}{Z_{21}}$$

Define:

$$\Delta_Z = Z_{11}Z_{22} - Z_{12}Z_{21}$$

Then:

$$AZ_L + B = \frac{Z_{11}Z_L + \Delta_Z}{Z_{21}}$$

and:

$$CZ_L + D = \frac{Z_L + Z_{22}}{Z_{21}}$$

Substitute these expressions into the ABCD efficiency equation.

4. Final single formula

The Match efficiency using only the Z-parameters and load impedance is:

$$\eta_{Match} = \frac{Re\{Z_L\}}{Re\left\{\left(\frac{Z_{11}Z_L + Z_{11}Z_{22} - Z_{12}Z_{21}}{Z_{21}}\right)\left(\frac{Z_L + Z_{22}}{Z_{21}}\right)^*\right\}}$$

This is the single, fully expanded formula.

A cleaner equivalent form is:

$$\eta_{Match} = \frac{Re\{Z_L\}}{Re\left\{\frac{(Z_{11}Z_L + Z_{11}Z_{22} - Z_{12}Z_{21})(Z_L + Z_{22})^*}{|Z_{21}|^2}\right\}}$$

Because $|Z_{21}|^2$ is real, it can also be written as:

$$\eta_{Match} = \frac{Re\{Z_L\}|Z_{21}|^2}{Re\{(Z_{11}Z_L + Z_{11}Z_{22} - Z_{12}Z_{21})(Z_L + Z_{22})^*\}}$$

Recommended final formula

For EVC software implementation, I recommend this version:

$$\eta_{Match} = \frac{Re\{Z_L\}|Z_{21}|^2}{Re\{[Z_{11}(Z_L + Z_{22}) - Z_{12}Z_{21}](Z_L + Z_{22})^*\}}$$

The output efficiency percentage is:

$$\eta_{Match}(\%) = 100 \frac{Re\{Z_L\}|Z_{21}|^2}{Re\{[Z_{11}(Z_L + Z_{22}) - Z_{12}Z_{21}](Z_L + Z_{22})^*\}}$$

5. Physical meaning behind the formula

From the Z-parameter equations:

$$V_2 = Z_{21}I_1 + Z_{22}I_2$$

Using:

$$I_2 = -I_L$$

and:

$$V_2 = Z_L I_L$$

we get:

$$Z_L I_L = Z_{21}I_1 - Z_{22}I_L$$

Therefore:

$$I_1 = \frac{Z_L + Z_{22}}{Z_{21}} I_L$$

Then:

$$V_1 = Z_{11}I_1 - Z_{12}I_L$$

Substitute the expression for I_1 :

$$V_1 = \left[\frac{Z_{11}(Z_L + Z_{22})}{Z_{21}} - Z_{12} \right] I_L$$

This can be rearranged to:

$$V_1 = \frac{Z_{11}(Z_L + Z_{22}) - Z_{12}Z_{21}}{Z_{21}} I_L$$

The final efficiency formula is simply:

$$\eta_{Match} = \frac{Re\{V_2 I_L^*\}}{Re\{V_1 I_1^*\}}$$

after cancelling $|I_L|^2$.